Expt.No: Date:

Linear Programming Problems – Constraint Optimization using Python

Problem 1

Question:

Maximize $Z = 3x + 2y$ subject to $x + y \leq 4$, $x \leq 2$, $y \leq 3$, $x,y \geq 0$

Aim:

To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP1", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 3*x + 2*y
lp += x + y <= 4
lp += x <= 2
lp += y <= 3
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
2.0 2.0 10.0
```

Problem 2

Question:

Minimize $Z = x + y$ subject to $2x + y \geq 6$, $x + 2y \geq 6$

Aim:

To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP2", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 5*x + 4*y
lp += 6*x + 4*y <= 24
lp += x + 2*y <= 6
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
3.0 1.5 21.0
```

Problem 3

Question:

Minimize $Z = x + y$ subject to $2x + y \geq 6$, $x + 2y \geq 6$

Aim:

To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP3", pulp.LpMinimize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += x + y
lp += 2*x + y >= 6
lp += x + 2*y >= 6
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
2.0 2.0 4.0
```

Problem 4

Question:

Maximize $Z = 4x + 3y$ subject to $2x + y \leq 8$, $x + 2y \leq 8$

Aim:

To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP4", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 4*x + 3*y
lp += 2*x + y <= 8
lp += x + 2*y <= 8
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
2.6666667 2.6666667 18.6666669
```

Problem 5

Question:

Minimize $Z = 3x + 5y$ subject to $x + y \geq 4$, $x \leq 3$, $y \leq 3$

Aim:

To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP5", pulp.LpMinimize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 3*x + 5*y
lp += x + y >= 4
lp += x <= 3
lp += y <= 3
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
3.0 1.0 14.0
```

Problem 6

Question:
Maximize $Z = 6x + 5y$ subject to $x + y \leq 5$, $3x + 2y \leq 12$

Aim:
To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP6", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 6*x + 5*y
lp += x + y <= 5
lp += 3*x + 2*y <= 12
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
2.0 3.0 27.0
```

Problem 7

Question:
Minimize $Z = 2x + 3y$ subject to $x + y \geq 5$, $x,y \geq 1$

Aim:
To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP7", pulp.LpMinimize)
x = pulp.LpVariable('x', lowBound=1)
y = pulp.LpVariable('y', lowBound=1)
lp += 2*x + 3*y
lp += x + y >= 5
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
4.0 1.0 11.0
```

Problem 8

Question:
Maximize $Z = 7x + 6y$ subject to $2x + y \leq 6$, $x + 2y \leq 6$

Aim:
To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP8", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 7*x + 6*y
lp += 2*x + y <= 6
lp += x + 2*y <= 6
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
2.0 2.0 26.0
```

Problem 9

Question:
Minimize $Z = x + 4y$ subject to $x + 2y \geq 8$, $x \geq 2$

Aim:
To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP9", pulp.LpMinimize)
x = pulp.LpVariable('x', lowBound=2)
y = pulp.LpVariable('y', lowBound=0)
lp += x + 4*y
lp += x + 2*y >= 8
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
8.0 0.0 8.0
```

Problem 10

Question:
Maximize $Z = 5x + 3y$ subject to $x + y \leq 7$, $x \leq 4$, $y \leq 5$

Aim:
To optimize the given objective function using Linear Programming.

Python Code:

```
import pulp
lp = pulp.LpProblem("LPP10", pulp.LpMaximize)
x = pulp.LpVariable('x', lowBound=0)
y = pulp.LpVariable('y', lowBound=0)
lp += 5*x + 3*y
lp += x + y <= 7
lp += x <= 4
lp += y <= 5
lp.solve()
print(x.value(), y.value(), pulp.value(lp.objective))
```

Output:

```
4.0 3.0 29.0
```